



Clinical Practice Guideline

Non-Tunneled and Tunneled Large Bore Central Venous Catheters

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to large bore central venous catheters (CVCs) including non-tunneled and tunneled CVCs. It ensures the delivery of a uniform, integrated multidisciplinary approach consistent with best practice.

The terms Non-Tunneled and Tunneled Large Bore CVCs are generic and can relate to different brands/types of catheters e.g., Hickman™, Vascath™. Please refer to procedural documentation for specific catheter details.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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Definitions

Central Venous Catheter (CVC)	Central venous access devices specifically designed to facilitate haemodialysis or apheresis. Note: In this document the term CVC refers to <u>large bore</u> CVCs only (>10French)
Non-Tunneled CVC	Non-tunneled CVC (e.g., Vascath TM): the catheter enters the vein (e.g., internal jugular vein) near the skin puncture site. The catheter is inserted by puncturing directly into the skin and vein without tunneling the subcutaneous tissue. Non-tunneled catheters are intended for short term use (≤ 3 weeks).
Tunneled CVC	Tunneled CVC (e.g., Hickman TM): tunneling adds a further barrier to the ingress of organisms. Tunneled CVCs are tunneled under the skin subcutaneously before entering a large vein (commonly the internal jugular vein) to reach the heart ⁵ . The skin exit site is distant from the venepuncture site to allow for optimal dressing and securement. Tunneled CVCs are designed for long term use (months to years).
Locking Agent	A medication that is placed into a CVC and left in situ whilst the device is not in use e.g., Taurolock TM , Heparin, Alteplase
Dacron cuff	If the CVC is cuffed, a short section of the subcutaneous line has a cuff bonded around the outside of the line. The cuff creates a further barrier to infection by surrounding tissue granulating into the cuff. If the cuffed line stays in place >3 weeks, the granulating of tissue into the cuff acts as a self-securement and seal for the line.

1.0 Management Pre and Post Insertion

1.1 Principles

Always observe the principles of Hand Hygiene and Aseptic Technique. Refer to SCGOPHCG Infection Prevention and Control Manual – Aseptic Technique policy.

Patient identification must be checked when matching patient's identity to care, therapy and services as per SCGOPHCG Patient Identification policy and SCGOPHCG Procedure Matching Policy

It is a legal requirement that all disciplines prescribing or administering medications comply with the Medicines and Poisons Act 2014, Medicines and Poisons Regulations 2016, Pharmacy Act 2010, and Hospital Guidelines.

It is a legislative requirement that a valid prescription exists before Schedule 4 and Schedule 8 medications are administered. Only valid prescriptions will be utilised by nursing staff to administer medication except for verbal orders and nurse-initiated medications, as per designated list.

SCGOPHCG recognises the role of carers and shall establish and maintain mechanisms for the involvement of carers in the provision of services.

1.2 Medical Requirements

Pre insertion procedure:

- Medical referral should detail the intended use of the CVC and any relevant medical information.

- Written informed consent is required prior to insertion of a CVC unless clinically indicated (e.g., emergency situations)
- Sedation should be prescribed if required.
- Pre-operative 2% nasal Mupirocin ointment and a 4% chlorhexidine wash should be prescribed as per the SCGOPHCG Mupirocin Nasal Ointment for Decolonisation Guideline^{1,2}
- Complete request for post insertion Chest X-ray. A chest X-ray is not required if CVC insertion is performed by interventional area in Radiology (hereafter called Interventional Radiology)
- Review current medication e.g., anticoagulation therapy may require cessation prior to procedure.
- Determine if prophylactic antibiotics are required.

Procedure:

Non-Tunneled CVC (e.g., VascathTM) insertion is performed using surgical aseptic technique. Procedure is usually performed by an anaesthetist in the designated areas of Post Anaesthetic Care Unit (PACU), Theatre, Radiology and Intensive Care Unit (ICU)³.

Tunneled CVC (e.g., HickmanTM) insertion is performed using surgical aseptic technique. Procedure is usually performed in Interventional Radiology.^{1,4}

Confirm with the medical team inserting CVC if the patient is required to fast pre-procedure.

Post Procedure:

Refer to the SCGOPHCG Guideline – Mupirocin Nasal Ointment for Decolonisation for details:

- Following an emergency procedure where the full decolonisation regimen cannot be completed prior to catheter insertion.
- Nasal mupirocin maintenance therapy following catheter insertion, and guidelines for continuation, for the life of the catheter.

The Operation Report should:

- Confirm correct tip placement¹.
- Confirm catheter patency.
- Specify type of catheter and number of lumens.
- Specify internal and external length of catheter.
- Detail any complications associated with catheter insertion in the Operation Record.
- State if 'Safe for use'.

1.3 Safety Information

Large bore central venous catheters should only be used for their intended treatment except in an emergency.

Chest X-Ray is necessary to exclude complications and confirm tip placement (unless femoral venous access used, or tunneled CVC is inserted under ultrasound guidance in Interventional Radiology).¹

CVC should not be used until placement of catheter is confirmed and documented as safe for use.

Medical and Nursing staff must confirm that catheter is safe to use. Accessing of Large

Bore Central venous catheters' is to be by staff proficient to do so i.e., within individual scope of practice.

Implement nursing care plan– Central Vascular Access Device Insertion Record.
In G65 Dialysis Unit: Update the electronic record e.g. Western Australian Nephrology Database (WAND).

Non-tunneled and tunneled large bore CVCs must be labelled to identify that a locking agent is in situ and must be withdrawn prior to use. Ensure catheter lock label is attached to dressing. (Labels are available from G65, G63 & Haematology Day ward).

Each lumen of the catheter must be clamped when not in use. A cap (positive pressure cap or, in the setting of dialysis, a high flow positive pressure valve e.g., Tego™ connector) should be attached to each lumen.

1.4 Nursing Management

- The management of both tunneled and non-tunneled large bore CVCs follows the same site assessment, line access, dressing requirement and flushing techniques.
- Refer to the SCGOPHCG Guideline – Mupirocin Nasal Ointment for Decolonisation.
- Inspect and palpate the insertion and exit site, and adjoining limb of any CVC each shift and prior to each access for signs of:
 - Infection (redness, heat, exudate)
 - Thrombus (redness, heat, swelling, pain)
 - Hematoma
 - Extravasation (redness, heat, swelling, pain when administering intravenous medication)
 - Bleeding
 - Catheter migration (visible cuff or excessive length of catheter protruding from skin)
 - Loose or soiled dressing or cap (see dressing and cap change)
 - Line / Tubing fracture (leaking)
 - Broken clamp (contact Intravascular Access CNC or Specialty CNM/ CNS for advice)
 - Medical device pressure injury caused by CVC components and/or dressing. Check CVC anchoring wings, lumens, hubs and caps are not exerting pressure into the skin, resulting in skin indentation.
- Document and report evidence of any complications to medical and senior nursing staff immediately.
- In case of excessive bleeding from insertion site/exit site, apply gauze pressure dressing over site and cover with transparent dressing. Direct manual pressure may be required. Notify medical team for review. Remove gauze pressure dressing after 24 hours and replace with usual dressing (see section 2.0).
- Suspected catheter fracture: apply clamp between the fracture and the exit site. Cover the fracture with sterile gauze and tape the area securely.
 - **OPH:** Immediately inform the Medical Officer/ Consultant and Senior Registered Nurse (CNM).
 - **SCGH:** Immediately inform the Medical Officer and Senior Registered Nurse (CNS/ CNM/ Afterhours CNS). Interventional Radiology may be able to salvage the catheter.

Weekdays In- hours: If further advice is needed contact the SCGH Intravenous Therapy Team.

- Document management using the Central Vascular Access Device Insertion Record (MR864.2), in/outpatient notes, nursing care plan and any relevant electronic records e.g., WAND.

Non-Tunneled CVC (e.g., VASCATH) TM

Renal: intended for haemodialysis treatment only, providing temporary access while long-term access is organised.

Haematology: intended for Peripheral Blood Stem Cell Harvesting (PBSCH), and plasmapheresis.

The device is relatively rigid and can cause significant patient discomfort during movement.

The catheter must be well anchored by sutures close to insertion site³.

Dressings must enhance device stability, patient comfort and maintain a barrier to infection. Cover CVC insertion site as per section 2.3

Tunneled CVC (e.g., HICKMAN) TM

Renal: intended for haemodialysis treatment only. They may also be used for subsequent intravenous supportive care under consultation with renal physician, G65 CNM, G63 CNS.

Haematology: intended for peripheral blood stem cell transplant (PBSCT) and plasmapheresis. They may also be used for subsequent intravenous supportive care under consultation with consultant Haematologist.

Surgical: used infrequently for outpatient delivery of home total parenteral nutrition (TPN).

The catheter is tunneled under the skin between an '**entry**' site and an '**exit**' site.

Entry site

The **entry** or insertion site is the point where the catheter enters the vein (usually near the clavicle). The catheter is not visible at this site.

- The wound at the **entry** site is sutured and dressed at the time of the procedure.
- Change entry site dressing after 7 days or prn. Dressing no longer required when suture is removed, and wound has healed.
- Remove sutures at the **entry** site 7 – 10 days after CVC insertion unless otherwise instructed by medical team.

Exit site

The **exit** site is the point where the catheter emerges through the skin (usually on chest wall).

- An anchoring suture is used at the **exit** site to close the skin around the catheter and to prevent dislodgement.
- During the period of Dacron cuff engraftment, the catheter should remain well anchored by sutures close to the exit site wound.
- Cover CVC exit site as per section 2.3
- Remove sutures at the **exit** site 21 days after catheter insertion unless otherwise instructed by medical team.

Educate patient regarding device care and management. Consider caregiver involvement as appropriate. Patient information brochure available: Tunneled large-bore central venous catheter (CVC) Patient Information (SCGOPHCG 15750821).

1.5 Procedural Information

For suture removal: See SCGOPHCG NPG Wound Management

2.0 Dressing and Cap Changes

2.1 Safety Information

Before removing exit site dressing, anchor the catheter below the insertion site with an adhesive tape (e.g., Micropore).

Advise patient to

- remain still during dressing change to avoid accidental dislodgement of catheter.
- face away from CVC and minimise talking to avoid contamination of wound.

Ensure catheter is clamped prior to cap change.

Ensure caution label is placed on top of the occlusive dressing when a locking agent is instilled. Labels are available from G65, G63 & Haematology Day ward.

2.2 Nursing Management

The following applies to the exit site dressing:

- Change dressing every 7 days or as required if the dressing becomes loose, wet, or soiled.⁴
- Change caps every 7 days or as required if contaminated.
 - If locking agent is used, withdraw 5mls of fluid and discard after cap change.
 - Flush catheter with 10mls 0.9% sodium chloride for injection after each cap change.
- Sterile gloves must be worn when changing dressing or caps.
- Document the dressing replacement and the condition of the insertion site in the patient's notes and on the Central Vascular Access Device Insertion Record.
- Long-term use of tunneled CVCs: where appropriate, patients may be educated to attend their own dressings (e.g., home TPN). Some patients choose to leave the site exposed when the site has healed.

2.3 Procedural Information

Requirements

- Dressing trolley
- Dressing pack
- Tegaderm CHG™ (chlorhexidine antimicrobial dressing),
OR Biopatch™ or Kendall antimicrobial foam disc^{2,3} with large, sterile, transparent, occlusive dressing (Opsite IV 3000™)
- Small, sterile, transparent, occlusive dressing (Tegaderm™)
- Sterile gloves x 2 packets
- Personal protective equipment (PPE)
- Chlorhexidine 2% in 70% alcohol solution⁵
- 2 x Chlorhexidine 2% and alcohol 70% wipes
- Adhesive tape (Micropore™)
- 3M Cavilon Barrier Wipe™

- Fixomul™: 2 lengths cut to 1.5cm x 10cm.
- 2 x 10ml syringes
- 2 x 19G drawing up needle.
- 2 x 10mL 0.9% sodium chloride for injection (or 2 x BD Posiflush™ Pre-filled Saline Syringes) (one for each lumen)⁶
- Scissors
- Caution label
- Wet strength disposal bag
- 2 x Cap e.g. Tego™ or positive pressure valve Max Plus™ (one for each lumen)

Procedure

1. Perform hand hygiene.
 2. Clean work surface with combined detergent disinfectant wipe. Allow to dry.
 3. Gather equipment for the procedure.
 4. Perform hand hygiene and prepare field.
 5. Open dressing pack and add equipment to sterile field using non-touch technique.
 6. Perform hand hygiene.
 7. Don non-sterile gloves.
 8. Anchor the catheter below the exit site with adhesive tape.
 9. Advise patient to remain still to avoid accidental dislodgment and to look away from CVC and minimise talking to avoid accidental contamination of wounds.
 10. Remove old transparent dressing. (NB anchoring tape in point 8 should remain to prevent movement of catheter).
 11. Dispose of used dressing in waste bag and remove non-sterile gloves.
 12. Perform hand hygiene and don sterile gloves.
 13. Place sterile sheet under catheter.
 14. Clean the catheter site (exit site) using 2% chlorhexidine and 70% alcohol solution and allow to air dry.¹ If patient is sensitive or has an allergy to chlorhexidine, Betadine solution may be used. Normal saline cleaning solution can be used to remove congealed or dried blood from catheter or exit site prior to cleaning with 2% chlorhexidine and 70% alcohol.
 15. Use barrier wipe for surrounding skin if necessary e.g., 3M Cavilon Barrier Wipe™.
 16. Dressing:
 - a. Place Tegaderm CHG™^{1,3,7} over catheter exit site **OR**
 - b. Place Biopatch™ antimicrobial dressing^{1,3,7} around catheter exit site with blue side up and white foam side next to patient's skin and apply sterile transparent occlusive dressing Opsite IV 3000™, **OR**
 - c. If patient is sensitive or has an allergy to chlorhexidine, Kendall antimicrobial disk^{1,3,7} may be used with a sterile transparent occlusive dressing Opsite IV 3000™ to secure.
- NB: the catheter should *not* be lifted to apply dressing, Biopatch or Kendall disc or to clean site as lifting the catheter may disturb wound healing and Dacron cuff engraftment. Check CVC anchoring wings, lumens, hubs and caps are not exerting pressure into the skin, resulting in skin indentation.
17. Place locking agent caution label on the dressing (Do not obscure exit site).
 18. Place small occlusive dressing over label.
 19. Remove anchor tape in point 8. Replace with new anchor tape (Fixomul™ 1.5 x 10cm) at base of occlusive dressing to anchor both lumens (Figure 1). Place a second anchor tape (Fixomul™ 1.5 x 10cm), sticky side down, beneath catheter at base of occlusive dressing, overlapping first anchor tape at sides (Figure 2).

Figure 1



Figure 2



20. Ensure lines are clamped.
21. Remove PPE. Perform hand hygiene.
22. Don sterile gloves.
23. Remove cap using a non-touch technique. Scrub the lumen with chlorhexidine 2% and alcohol 70% wipes for 30 seconds^{1,4} using different areas of the wipe. Allow to air dry. If patient is sensitive or has an allergy to chlorhexidine, use 70% alcohol wipes.
24. Connect new cap.
25. Unclamp line.
26. Attach 10ml syringe, withdraw 5ml fluid and discard (to remove locking agent and air from cap).
27. Flush catheter with 10ml BD Posiflush™ 0.9% sodium chloride prefilled syringe⁶ using a rapid, pulsating, push/pause technique⁸.
28. Clamp line.
29. Repeat for other lumens.
30. Dispose of waste.
31. Remove PPE. Perform hand hygiene.
32. Clean the work surface with combined detergent disinfectant wipe. Allow to dry.
33. Perform hand hygiene.
34. Document change of dressing and caps and "next due" date on the Central Vascular Access Device Insertion Record (MR 864.2).

3.0 Accessing, Flushing and Locking

3.1 Medical Requirements

Inpatient use

If in use, catheter locking agent to be prescribed by the Medical Officer in the 'Regular medications' section on medication chart **to the volume of each lumen + 0.1ml** stating "post IV medications" (e.g., intravenous antibiotics). The volume of each lumen is written on the lumen or clamp of the catheter.

Locking agents:

- Heparinised saline 50 units/5ml when IV treatment regime <8 hourly
- OR
- Heparin 1000 units/ml when IV treatment ≥ 8 hourly

Patients at high risk, with a history of catheter related blood stream infections or contraindications: consultation with Infectious Diseases in conjunction with the patient's Renal, Haematology or Surgical specialist for consideration of alternative locking agent recommended.

When patient's IV treatment is completed, return to usual locking agent regime (see

outpatient use).

Flushing catheters with a rapid, pulsatile, push-pause method produces turbulent flow which may decrease formation of clots on catheter wall and improve flow⁸.

Locking agent replacement is required every 7 days.

Dormant CVCs must be flushed weekly.

Outpatient use

Catheter locking agent to be prescribed by the Medical Officer in the 'Once only, pre-medication & nurse/midwife-initiated medicines' section on the medication chart **to the volume of each lumen + 0.1ml**.

G65, Dialysis (SCGH): Catheter locking agent will be prescribed on the Nurse Initiated Medication Approval Form (NIMA).

- Taurolidine 1% Sodium Citrate 4% Heparin 500unit (ampoule)^{1,2,3,9}
OR
- Heparin 5000units/mL⁹

The preferred locking agent is taurolidine/citrate/heparin (Taurolock HEP500).

Haematology Day Unit: Catheter locking agent will be prescribed on the NIMA.

- Taurolidine 1% Sodium Citrate 4% Heparin 500unit (5ml ampoule)^{1,2,3,9}

All patients should have a catheter locking agent in situ for the first six weeks post catheter insertion unless medically contra-indicated.⁴ Patients requiring CVC for haemodialysis will have locking agent in situ for life of catheter.

Patients at high risk, with a history of catheter related blood stream infections or contraindications: consultation with Infectious Diseases in conjunction with the patient's Renal, Haematology or Surgical specialist for consideration of alternative locking agent recommended.^{1,3,11}

Locking agent replacement required every 7 days (minimum)⁴.

3.2 Safety Information

CVCs should only be used for their intended treatment except in an emergency.

Staff should not access CVC if not within their scope of practice.

If a patient is admitted with a tunneled large bore CVC inserted at another hospital, a chest X-ray should be performed to confirm correct tip placement prior to use¹.

If catheter migration is suspected inform Medical Officer. Chest X-ray may be required to confirm correct catheter tip position and device safety.

Locking agent: instill lumen volume + 0.1ml. Volume is written on the lumen or clamp of the catheter.

If the catheter contains a locking agent - do not flush.

Remove catheter locking agent by withdrawing 5mls of fluid prior to use.

Use Luer lock syringes for flushing large bore CVC: minimum size 10mls to avoid catheter damage or rupture.

3.3 Nursing Management

- Large bore permanent or temporary CVCs should not be used for routine access e.g., administration of intravenous fluids, unless approved by medical team or in an emergency situation.
- Flush with 10mls 0.9% sodium chloride after each catheter access and cap change to maintain patency.
- Instil locking agent post flush if prescribed.
- When a tunneled CVC is used during an inpatient admission for treatment and/or supportive therapy:
 - Confirm correct catheter placement prior to each use by obtaining a brisk blood return on aspiration.
 - Flush lumens with 10mls 0.9% sodium chloride using a rapid, pulsating, push/pause technique immediately before and after use⁶.
 - Ensure all traces of blood and medication are cleared from the positive pressure valve and lumens (NB more than 10 ml may be required to achieve this⁸).
 - After 0.9% sodium chloride flush, disconnect syringe from cap (positive pressure valve) and clamp lumen.
 - If not in frequent use (i.e., $\leq$ weekly) and on discharge, discuss the need for an alternative locking agent with medical team.

3.4 Procedural Information

Requirements

- Dressing pack
- 2 x 10ml syringe
- 2 x 19G drawing up needle.
- 2 x 10mL 0.9% sodium chloride for injection (or 2 x BD Posiflush™ Pre-filled Saline Syringes)
- 2 x 3ml syringes
- Prescribed catheter locking agent.
- 2 x large 70% alcohol-impregnated wipes
- Non-Sterile gloves & other PPE
- Dressing trolley

Procedure

1. Perform hand hygiene.
2. Clean work surface with combined detergent disinfectant wipe. Allow to dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare field.
5. Open dressing pack and add equipment to sterile field using non-touch technique.
6. Unclamp line.
7. Perform hand hygiene.
8. Don non-sterile gloves and other PPE as indicated.
9. Place sterile sheet under catheter.
10. Scrub the cap⁴ using a non-touch technique with a large 70% alcohol-impregnated wipe for 30 seconds using different areas of the wipe ⁶. Allow to air dry.
11. Attach 10ml syringe using a non-touch technique.
12. Withdraw 5ml of fluid to remove catheter locking agent.
13. Flush with 10ml 0.9% sodium chloride.
14. Administer treatment via syringe or infusion pump as indicated.
15. Flush with 0.9% sodium chloride 50ml bag post infusion therapy.
16. Flush with 10ml 0.9% sodium chloride using a rapid, pulsating, push/pause

technique⁸.

17. Prepare catheter locking agent in 3ml syringe (1 x 3ml syringe per lumen).
18. Administer catheter locking agent using a constant, even force on syringe plunger to prevent backflow of blood⁸.
19. Clamp line.
20. Attach locking agent caution label.
21. Dispose of waste.
22. Remove PPE. Perform hand hygiene.
23. Clean work surface with a combined detergent disinfectant wipe.
24. Perform hand hygiene.

4.0 Blood Sampling

4.1 Medical Requirements

Pathology blood sample form should clearly state that blood samples are obtained from CVC (e.g., Hickman™ or Vascath™).

4.2 Safety Information

Routine blood sampling is not recommended due to the increased risk of infection and catheter occlusion. Blood samples should be taken from peripheral veins whenever possible. However, in event of poor peripheral access, blood sampling from CVC may be necessary.

Blood samples may be obtained from a large bore catheter except when vasoactive medicines, chemotherapy, total parenteral nutrition, or heparin is being infused. **Do not** cease these infusions for blood sampling; rather use an alternative route.

4.3 Nursing Management

Ensure catheter lumens are always clamped when not in use.

4.4 Procedural Information

Requirements

- Dressing pack
- 1 x 10ml syringe
- 1 x 19G drawing up needle.
- 2 x 10mL 0.9% sodium chloride for injection (or 2 x BD Posiflush™ Pre-filled Saline Syringes)
- 1 x 3ml syringe
- Prescribed locking agent
- 1 x 20ml syringe or Vacutainer blue top
- Vacutainer 'pink' or 'blue' top
- Blood specimen tubes
- Large 70% alcohol-impregnated wipe
- Nonsterile gloves and other PPE
- Dressing trolley

Procedure

1. Perform hand hygiene.
2. Clean work surface with combined detergent disinfectant wipe. Allow to dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene.
5. Turn off infusions one minute prior to sampling if possible.
6. Perform hand hygiene.

7. Open dressing pack and add equipment to sterile field using non-touch technique.
8. Perform hand hygiene.
9. Place sterile sheet under catheter.
10. Don nonsterile gloves and other PPE.
11. Place sterile sheet under catheter.
12. Scrub the cap and lumen using a non-touch technique with a large 70% alcohol-impregnated wipe for 30 seconds⁴ using different areas of the wipe. Allow to air dry.
13. Attach 10ml syringe using a non-touch technique and unclamp lumen.
14. Withdraw 5ml lock solution.
 - a) standard blood sampling: discard.
 - b) blood cultures: do not discard (unless antibiotic or antimicrobial locked).
15. Withdraw required sample using vacutainer system or 20ml syringe. (Withdraw sample slowly to decrease risk of haemolysis of sample).
16. Flush with 2 x 10ml 0.9% sodium chloride using a rapid, pulsating, push/pause technique⁸. Remove syringe.
17. If prescribed, prepare catheter locking agent in 3ml syringe.
18. If prescribed, administer catheter locking agent using a constant, even force on syringe plunger to prevent backflow of blood⁸.
19. Clamp line (Refer to section 3: Accessing, Flushing, and Locking).
20. If sampling using a 20ml syringe, attach syringe to pink vacutainer. Transfer blood to required specimen tubes.
21. Dispose of waste.
22. Remove PPE. Perform hand hygiene.
23. Clean work surface with combined detergent disinfectant wipe.
24. Perform hand hygiene.

5.0 Unblocking Lumens using Alteplase

5.1 Medical Requirements

Alteplase must be prescribed on the medication chart by an authorised prescriber.

In inpatient areas: Alteplase is prescribed as a once only medication as per SCGOPHCG Medicines Management Policy.

G65, Dialysis: Alteplase is prescribed as PRN medication as per SCGOPHCG Medicines Management Policy.

5.2 Safety Information

Alteplase is indicated for occluded catheters to promote patency of the CVC and to resolve clotting.^{4,10.}

Alteplase 2mg in 2mls is supplied as a frozen, pre-filled syringe and must be defrosted prior to use.

Do **not** use Alteplase:

- If catheter-associated infection is suspected.
- To unblock lumens being used for infusions containing inotropic drugs or vasoactive drugs. Lines with inotropic or vasoactive drugs are not routinely flushed hence use of Alteplase may unintentionally expose patients to inotropic or vasoactive drugs left in line.

- If unable to aspirate or inject any fluid into the line, there is no indication for Alteplase. Contact the Intravascular Access CNC or Specialty CNM/CNS for advice.

Potential complications of use include:

- Increased anticoagulant effect.
- Haemorrhage NB: any Alteplase released into the circulation is metabolised rapidly by the liver (plasma half-life = 5 mins) so risk of bleeding is minimal.
- Anaphylaxis (rare < 2%).

5.3 Nursing Management

- Exclude other possible causes of occlusion e.g., kinked catheter, incorrect position, clamp engaged, catheter migration.
- Alteplase requires double, independent checking and verification as per NPG # 4 Intravenous Therapy
- Dosage is determined by the volume of lumen. Volume is written on the lumen or clamp of the catheter.
 - Unblocking a single lumen of a multiple lumen catheter:
Administer Alteplase into the affected lumen to volume of the lumen + 0.1ml.
 - Unblocking more than one lumen of a multiple lumen catheter:
Administer Alteplase into each affected lumen to volume of the lumen + 0.1ml.

If the lumen volume is greater than 2mls, further dilute solution with sterile 0.9% sodium chloride for injection to the desired volume.

Inpatient areas: repeat administration of Alteplase within a 24-hour period if lumen patency remains poor.

G65, Dialysis (SCGH): follow Algorithm for the use of Thrombolytic Agent in Occluded Tunnelled Haemodialysis Catheter W:\Renal_Dialysis\SCG\Dialysis-Nursing\HD\Departmental Clinical Guidelines\AA-Alteplase Algorithm version 4.pdf.

5.4 Procedural Information

Part 1: Preparation and Instillation of Alteplase

Requirements:

- Dressing pack
- 2 x 10ml Luer lock syringes for discard
- 1 x 19G drawing up needle (2 required if more than one lumen involved)
- 2 x 10mL 0.9% sodium chloride for injection (or 2 x BD Posiflush™ Pre-filled Saline Syringes)
- 1 x Alteplase 2mg in 2ml prefilled syringe per occluded lumen
- 1 x 0.9% sodium chloride ampoule (if lumen priming volume exceeds 2ml)
- 2 x Large 70% alcohol wipe
- PPE
- Nonsterile gloves and other PPE
- 1 x locking agent caution label
- Dressing trolley

Procedure:

1. Perform hand hygiene.
2. Clean work surface with combined detergent disinfectant wipe.
3. Gather equipment for the procedure.

4. Perform hand hygiene.
5. Required Alteplase volume = CVC lumen volume + 0.1 ml. For each lumen requiring treatment, prepare Alteplase 2mg in 2ml prefilled syringe by discarding surplus medication. If lumen volume is greater than 2mls, dilute with 0.9% sodium chloride for injection to required volume.
6. Open dressing pack and add equipment to sterile field using non-touch technique.
7. Unclamp line.
8. Perform hand hygiene.
9. Don non-sterile gloves and other PPE as indicated.
10. Place sterile sheet under catheter.
11. Scrub the cap and lumen using a non-touch technique⁴ with a large 70% alcohol-impregnated wipe for 30 seconds using different areas of the wipe. Allow to air dry.
12. Attach 10ml syringe using a non-touch technique.
13. Withdraw a minimum of 5mls of fluid.
14. Flush with 0.9% sodium chloride using a rapid, pulsating, push/pause technique⁸. Remove syringe.
14. Attach Alteplase pre-filled syringe using non-touch technique.
15. Instil required dose of Alteplase into the blocked lumen via attached cap using a constant, even force on syringe plunger to prevent backflow of blood⁸.
15. Clamp line.
16. Attach locking agent caution label.
17. Repeat in all affected lumens.
18. Dispose of waste.
19. Remove PPE. Perform hand hygiene.
20. Clean the work surface with combined detergent disinfectant wipe.
21. Perform hand hygiene.
22. Document date and time of instillation on WA Hospital Medication Chart and document in the patient notes.
23. Allow Alteplase to dwell in lumen for a minimum of 1 hour before attempting aspiration. Ideally leave for 4 hours or, optimally, for 24 hours or until the next outpatient treatment.

Part 2: Aspiration of Alteplase

Requirements

- Dressing pack
- 10ml Luer lock syringes for discard
- 2 x 10mL 0.9% sodium chloride for injection (or 2 x BD Posiflush™ Pre-filled Saline Syringes)
- Large 70% alcohol-impregnated wipe
- Nonsterile gloves and other PPE
- Dressing trolley

Procedure

1. Perform hand hygiene.
2. Clean work surface with combined detergent disinfectant wipe.
3. Gather equipment for procedure.
4. Perform hand hygiene.
5. Unclamp line.
6. Perform hand hygiene.
7. Don non-sterile gloves and other PPE as indicated.
8. Place sterile sheet under catheter.
9. Scrub the cap and lumen using a non-touch technique with a large 70% alcohol-

- impregnated wipe for 30 seconds⁴ using different areas of the wipe. Allow to air dry.
10. Attempt to aspirate Alteplase from lumen. If unable to aspirate, Alteplase may be safely flushed into lumen.
 11. Once occlusion is cleared, flush well with 10-20mls sodium chloride 0.9% using a rapid, pulsating, push/pause technique⁸.
 12. Following establishment of catheter function, lock with usual locking agent as per area policy.
 13. Dispose of waste.
 14. Remove PPE. Perform hand hygiene.
 15. Clean work surface with combined detergent/ disinfectant wipe.
 16. Perform hand hygiene.
 17. Document in patient notes.
 18. A second dose of Alteplase may be administered within a 24-hour period if catheter function is not restored after the first dose.
 19. If the catheter remains occluded, contact Intravascular Access CNC or Specialty CNM/ CNS for advice.

6.0 Removal of non-tunneled large bore CVC

6.1 Medical Requirements

TUNNELED CVC removal is to be performed by medical staff experienced in the procedure.

A valid medical team request is required for the removal of a non- tunneled catheter.

6.2 Safety Information

In event of accidental arterial placement of CVC: **DO NOT REMOVE**. Consult with Vascular Surgery and Interventional Radiology.

Ensure patient's coagulation profile is known and reviewed by medical staff prior to catheter removal. Ideally, the platelet count should be more than $50 \times 10^9/l$ and the INR less than 1.5 before catheter CVC removal. However, clinician's discretion is required for each individual case. *

Catheter **should not** be removed for 6 hours after heparinized haemodialysis.

Caution should be used in the removal of large bore CVCs to reduce risk of bleeding and air embolism:

- Position the patient supine for catheter removal.
- Apply digital pressure until haemostasis achieved using manual compression and or adjunct approaches e.g., haemostatic agents designed to potentiate clot formation.
- Apply sterile gauze and transparent dressing to the access site to seal the skin-to-vein tract and decrease the risk of air embolus.

6.3 Nursing Management

Removal of TUNNELED catheter is to be performed by medical staff only.

G65 Dialysis (SCGH): refer to G65 area specific nursing guideline No.11. Removal of Tunneled Central Venous Catheters on G65.

CVC must be removed prior to discharge, unless indicated for further outpatient treatment. (Refer to Departmental Clinical Guidelines for Renal and Haematology patients).

Reason for CVC removal should be documented on Central Vascular Access Device Insertion Record & form faxed to # indicated on form.

CVC should only be replaced if clinically indicated.

If CVC is being removed for presumed catheter related blood stream infection, catheter tip is sent for culture.

6.4 Procedural Information

Requirements

- Dressing pack
- Sterile gloves
- Nonsterile gloves and other PPE
- Stitch cutter.
- Sterile specimen container
- Sterile scissors (if catheter tip is to be sent to microbiology)
- Sterile gauze
- Chlorhexidine 2% in 70% alcohol swab stick or solution
- Occlusive dressing +/- Calcium Alginate dressing
- Dressing trolley

Procedure

1. Perform hand hygiene.
2. Clean work surface with combined detergent disinfectant wipe.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare field.
5. Open dressing pack and add equipment to sterile field using non-touch technique.
6. Place patient in supine position.
7. Perform hand hygiene.
8. Don nonsterile gloves and other PPE.
9. Remove CVC dressing.
10. Perform hand hygiene.
11. Don sterile gloves.
12. Place sterile sheet under catheter.
13. Clean entry site using 2% chlorhexidine in 70% alcohol swab stick or solution. Allow to air dry.⁵
14. Remove sutures.
15. Instruct patient to perform Valsalva manoeuvre (take 2 deep breaths and hold on the third breath).
16. Remove the catheter in a deliberate, fluid and constant motion.
17. Apply sterile gauze, with firm pressure over the exit site.
18. Instruct patient to breathe once gauze and pressure applied.
19. Maintain pressure for minimum 5 minutes, or until bleeding ceases. A calcium alginate dressing (e.g., Kaltostat™) may promote haemostasis.
20. Apply an occlusive dressing over the exit site. Leave intact for 48 hours.
21. Inspect the catheter to confirm tip is intact (not fractured).
22. If sepsis is suspected, collect blood cultures (peripheral and via line) immediately prior to catheter removal. Send approximately 2cm tip of CVC to microbiology with pathology request form.
23. Dispose of waste.

24. Remove PPE. Perform hand hygiene.
25. Clean work surface with combined detergent/ disinfectant wipe.
26. Perform hand hygiene.
27. Observe site each shift for signs of bleeding or swelling. If bleeding occurs replace with a calcium alginate dressing using standard aseptic technique.
28. Document catheter removal in the patient's record and on Central Vascular Access Device Insertion Record. In G65 Dialysis Unit: update WAND clinical record.

7.0 Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Carer's Recognition Act 2004 (WA).
- Work Health and Safety Act 2020
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010

Policy:

- NMHS Consent to Treatment Policy
- NMHS Clinical Documentation Policy
- SCGOPHCG Patient Identification Policy
- SCGOPHCG Procedure Matching Policy
- SCGOPHCG Infection Prevention and Management Aseptic Technique Policy
- WA Department of Health Mandatory Policy 0131/20 High Risk Medication Policy.

Guideline:

- SCGOPHCG Medication Management Guideline – Mupirocin Nasal Ointment for Decolonisation
- SCGOPHCG Nursing Practice Guideline Wound Management
- SCGOPHCG Intravenous Therapy

8.0 References

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‡Expert Opinion of Doctor Aron Chakera Renal Physician (cited October 2024)

9.0 Authorisation & Version Control

Policy Sponsor:						Nurse Co-Director Cancer Imaging Clinical Service						
Policy Contact:						CNC IV Access						
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Applicable Standards			NSQHS Std 1: Clinical Governance  Std 2: Partnering with Consumers  Std 3: Preventing and Controlling Healthcare Associated Infection  Std 4: Medication Safety  Std 5: Comprehensive Care  Std 6: Communicating for Safety  Std 8: Recognising and Responding to Acute Deterioration 									
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